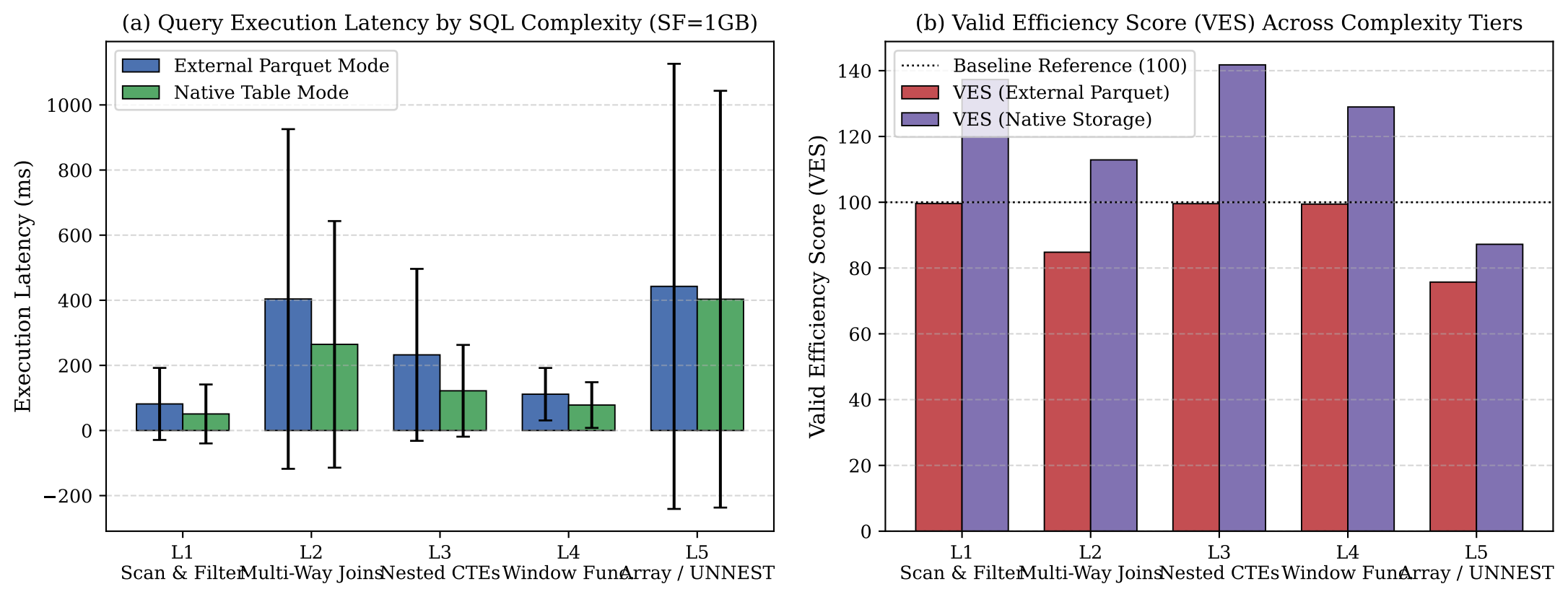
(a) Query Execution Latency by SQL Complexity (SF=1GB)
External Parquet Mode
Native Table Mode
Execution Latency (ms)
1000
800
600
400
200
0
−200
L1
Scan & Filter
L2
Multi-Way Joins
L3
Nested CTEs
L4
Window Funcs
L5
Array / UNNEST
(b) Valid Efficiency Score (VES) Across Complexity Tiers
Baseline Reference (100)
VES (External Parquet)
VES (Native Storage)
Valid Efficiency Score (VES)
140
120
100
80
60
40
20
0
L1
Scan & Filter
L2
Multi-Way Joins
L3
Nested CTEs
L4
Window Funcs
L5
Array / UNNEST